ECE 332: Engineering Electromagnetics II

Lecture 9

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Current Flow in a Good Conductor (Continued)

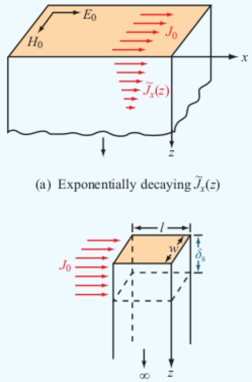
- The **total current passing through a slice of the conductor with width w in the y direction** is

$$\tilde{I} = \frac{J_0 w \delta_s}{1+j}. \qquad (1)$$

  This is an excellent approximation if the depth is finite $>$ a few times δ_s. (This is from Lecture 7.)

- If the slice has thickness ℓ in the x direction, the **voltage across the slice at the surface** is

$$\tilde{V} = E_0 \ell = \frac{J_0}{\sigma}\ell. \qquad (2)$$



(a) Exponentially decaying $\tilde{J}_x(z)$

(b) Equivalent J_0 over skin depth δ_s

Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

- The **impedance of the slice** is

$$Z = \frac{\tilde{V}}{\tilde{I}} = Z_s \frac{\ell}{w} \ (\Omega), \qquad (3)$$

where

$$Z_s = \frac{1+j}{\sigma \delta_s} \qquad (4)$$

is the **internal** or **surface impedance**.

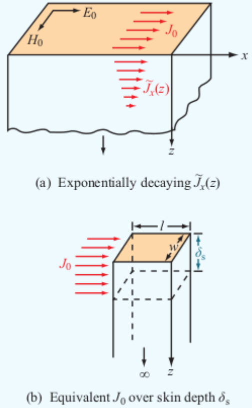


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

- Here,

$$Z_s = R_s + j\omega L_s \qquad (5)$$

with **surface resistance**

$$R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \ (\Omega) \qquad (6)$$

and inductance

$$L_s = \frac{1}{2}\sqrt{\frac{\mu}{\pi f \sigma}} \ (\mathrm{H}). \qquad (7)$$

- The **ac resistance of a slab of width** w **and length** ℓ is

$$R = R_s \frac{\ell}{w}. \qquad (8)$$

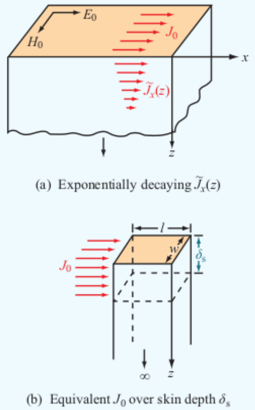


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

Quick Recap (Continued)

- The **ac resistance of a coaxial cable per unit length** is

$$R' = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right) \ (\Omega/\mathrm{m}), \qquad (9)$$

where R_s is the surface resistance of the conducting material, a is the radius of the inner conductor, b is the radius of the outer conductor, and both conductors are at least several times the skin depth. (The surface resistance and skin depth depend on the frequency of the wave.)

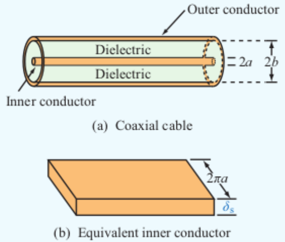


Figure 7-16 The inner conductor of the coaxial cable in (a) is represented in (b) by a planar conductor of width $2\pi a$ and depth δ_s, as if its skin has been cut along its length on the bottom side and then unfurled into a planar geometry.

Quick Recap (Continued)

Electromagnetic Power Density

- The **Poynting vector** is

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\mathrm{W/m^2}), \tag{10}$$

specifies the instantaneous power of an EM wave in the direction $\hat{\mathbf{k}}$ the wave propagates.

- The **power flowing through an area A with unit surface vector $\hat{\mathbf{n}}$** is

$$P = \int_A \mathbf{S} \cdot \hat{\mathbf{n}}\, dA = SA\cos\theta \quad (\mathrm{W}), \tag{11}$$

where $S = |\mathbf{S}|$ and θ is the angle between $\hat{\mathbf{k}}$ and $\hat{\mathbf{n}}$.

- The **average power density**

$$\mathbf{S}_{\mathrm{av}} = \frac{1}{2}\mathrm{Re}\left\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\right\} \quad (\mathrm{W/m^2}) \tag{12}$$

is a better measure of the power the wave carries.

Quick Recap (Continued)

- In a **lossless medium**,

$$\mathbf{S}_{\text{av}} = \frac{1}{2\eta}|\tilde{\mathbf{E}}|^2\hat{\mathbf{k}} \quad (\text{W/m}^2). \tag{13}$$

- In a **lossy medium**,

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta\hat{\mathbf{z}} \quad (\text{W/m}^2), \tag{14}$$

  where

$$\eta_c = |\eta_c|e^{j\theta_\eta}. \tag{15}$$

- In a lossy medium, the **power decreases at twice the rate** that the electric and magnetic fields attenuate.
- A power ratio P_1/P_2 is often expressed in **decibels**:

$$G\ (\text{dB}) = 10\log\left(\frac{P_1}{P_2}\right) = 20\log\left(\frac{V_1}{V_2}\right). \tag{16}$$

Quick Recap (Continued)

Here, V_1 and V_2 are voltages applied across the same resistor and P_1 and P_2 are the power dissipated in each case, respectively.

- Decibels are **usually used** to compare a **measured power** P_{meas} to a **reference power** P_0.
- The rate of decrease of $S_{\text{av}}(z)$ is the **attenuation rate**

$$A = 10 \log \left(\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)} \right) = -8.69 \left(\alpha \ (\text{Np/m}) \right) z = -\left(\alpha \ (\text{dB/m}) \right) z, \tag{17}$$

where

$$\alpha \ (\text{dB/m}) \equiv 8.69 \alpha \ (\text{Np/m}). \tag{18}$$

- The **attenuation rate** can be written in terms of the **magnitude of the electric field**

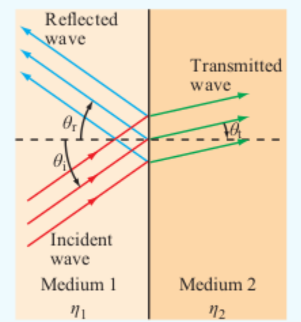
$$A = 20 \log \left(\frac{|E(z)|}{|E(0)|} \right) \quad (\text{dB}). \tag{19}$$

Quick Recap (Continued)

Chapter 8: Wave Reflection and Transmission

Suppose Medium 1 and Medium 2, with different electromagnetic properties, meet and that an EM wave traveling in Medium 1 reaches the boundary. Then

- A portion of the wave is **transmitted** into Medium 2.
- Another portion of the wave is **reflected** back into Medium 1.
- The **direction of propagation of the transmitted wave is different** from the original direction of propagation.
- The **speed of propagation changes** in Medium 2.

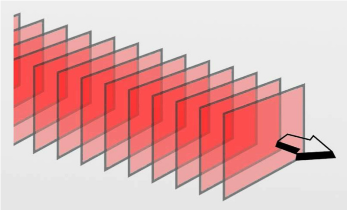


(b) Ray representation of oblique incidence

Quick Recap (Continued)

Some useful terminology:

- A **ray** is a line in the direction $\hat{\mathbf{k}}$ that an EM wave travels.
- A **wavefront** is a surface where the phase of a wave is constant.
- The wave traveling toward the boundary is the **incident wave**.
- The wave transmitted through the boundary is the **transmitted wave**.
- The wave reflected back into Medium 1 is the **reflected wave**.



Quick Recap (Continued)

- If **direction of travel** of the incident wave is **normal to the boundary**, then the reflected and transmitted waves travel normal to the boundary.

- This is analogous to what occurs at the boundary between two **transmission lines**.

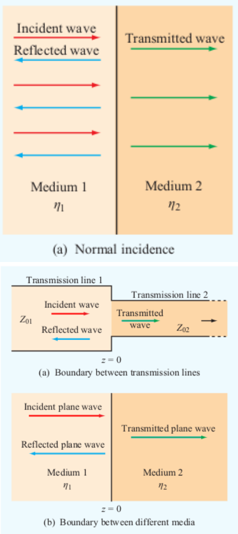


Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Quick Recap (Continued)

Suppose both media are **lossless** and that the **incident wave travels in the direction of the normal to the boundary**.

Medium 1

- Fills half-space $z \leq 0$
- Permittivity ϵ_1
- Permeability μ_1
- $k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $\eta_1 = \sqrt{\mu_1/\epsilon_1}$

Medium 2

- Fills half-space $z \geq 0$
- Permittivity ϵ_2
- Permeability μ_2
- $k_2 = \omega\sqrt{\mu_2\epsilon_2}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$

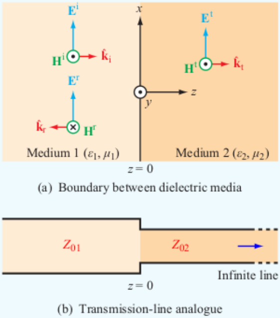


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

<u>Incident wave</u>: x-polarized and traveling in direction $\hat{\mathbf{k}}_i = \hat{\mathbf{z}}$.

Quick Recap (Continued)

Incident Wave ($\mathbf{E}^i, \mathbf{H}^i$)

$$\tilde{\mathbf{E}}^i(z) = E_0^i e^{-jk_1 z}\hat{\mathbf{x}} \tag{20}$$
$$\tilde{\mathbf{H}}^i(z) = (E_0^i/\eta_1)e^{-jk_1 z}\hat{\mathbf{y}} \tag{21}$$

Reflected Wave ($\mathbf{E}^r, \mathbf{H}^r$)

$$\tilde{\mathbf{E}}^r(z) = E_0^r e^{jk_1 z}\hat{\mathbf{x}} \tag{22}$$
$$\tilde{\mathbf{H}}^r(z) = -(E_0^r/\eta_1)e^{jk_1 z}\hat{\mathbf{y}} \tag{23}$$

Transmitted Wave ($\mathbf{E}^t, \mathbf{H}^t$)

$$\tilde{\mathbf{E}}^t(z) = E_0^t e^{-jk_2 z}\hat{\mathbf{x}} \tag{24}$$
$$\tilde{\mathbf{H}}^t(z) = (E_0^t/\eta_2)e^{-jk_2 z}\hat{\mathbf{y}} \tag{25}$$

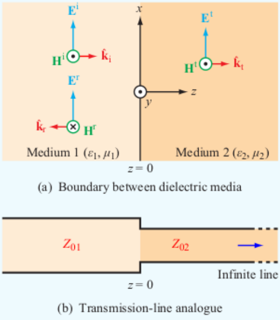


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

E_0^i, E_0^r, and E_0^t are the **amplitudes** of $\mathbf{E}^i$, $\mathbf{E}^r$, and $\mathbf{E}^t$ at $z = 0$, resp.

Quick Recap (Continued)

- The **tangential boundary conditions** imply

$$E_0^r = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}\right) E_0^i = \Gamma E_0^i, \quad E_0^t = \left(\frac{2\eta_2}{\eta_2 + \eta_1}\right) E_0^i = \tau E_0^i, \tag{26}$$

where the **reflection coefficient** is

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \tag{27}$$

and the **transmission coefficient** is

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 1 + \Gamma. \tag{28}$$

- When the media are **lossless**, Γ and τ are **real-valued**.
- **Nonmagnetic media**: $\mu_1 = \mu_2 = \mu_0$

$$\eta_1 = \eta_0/\sqrt{\epsilon_{r1}}, \quad \eta_2 = \eta_0/\sqrt{\epsilon_{r2}}, \quad \text{and} \quad \Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}}. \tag{29}$$

The analogous transmission-line configuration is a junction where a **lossless line (Line 1)** with characteristic impedance Z_{01} meets an infinitely long **lossless line (Line 2)** with characteristic impedance Z_{02} at $z = 0$. The **voltage reflection coefficient** at $z = 0$ going from Line 1 to Line 2 is

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} \qquad (30)$$

In fact, the corresponding transmission-line quantities $(\tilde{V}, \tilde{I}, \beta, Z_0)$ and plane-wave quantities $(\tilde{E}, \tilde{H}, k, \eta)$ **satisfy the same equations**.

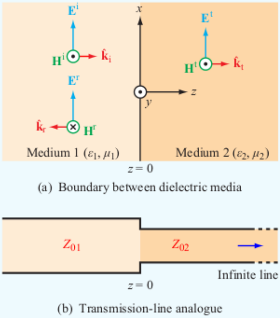


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

That means that you can apply the same techniques and understanding developed in Chapter 2 for transmission lines to the case of EM wave reflection and transmission in lossless media when the incident wave is traveling in the normal direction to the boundary.

Table 8-1 **Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.**

Plane Wave [Fig. 8-4(a)]		Transmission Line [Fig. 8-4(b)]	
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}}E_0^{\mathrm{i}}(e^{-jk_1z}+\Gamma e^{jk_1z})$	(8.11a)	$\widetilde{V}_1(z) = V_0^+(e^{-j\beta_1z}+\Gamma e^{j\beta_1z})$	(8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}}\,\dfrac{E_0^{\mathrm{i}}}{\eta_1}(e^{-jk_1z}-\Gamma e^{jk_1z})$	(8.12a)	$\widetilde{I}_1(z) = \dfrac{V_0^+}{Z_{01}}(e^{-j\beta_1z}-\Gamma e^{j\beta_1z})$	(8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}}\tau E_0^{\mathrm{i}}e^{-jk_2z}$	(8.13a)	$\widetilde{V}_2(z) = \tau V_0^+e^{-j\beta_2z}$	(8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}}\tau\,\dfrac{E_0^{\mathrm{i}}}{\eta_2}\,e^{-jk_2z}$	(8.14a)	$\widetilde{I}_2(z) = \tau\,\dfrac{V_0^+}{Z_{02}}\,e^{-j\beta_2z}$	(8.14b)
$\Gamma = (\eta_2-\eta_1)/(\eta_2+\eta_1)$		$\Gamma = (Z_{02}-Z_{01})/(Z_{02}+Z_{01})$	
$\tau = 1+\Gamma$		$\tau = 1+\Gamma$	
$k_1 = \omega\sqrt{\mu_1\varepsilon_1}\,, \quad k_2 = \omega\sqrt{\mu_2\varepsilon_2}$		$\beta_1 = \omega\sqrt{\mu_1\varepsilon_1}\,, \quad \beta_2 = \omega\sqrt{\mu_2\varepsilon_2}$	
$\eta_1 = \sqrt{\mu_1/\varepsilon_1}\,, \quad \eta_2 = \sqrt{\mu_2/\varepsilon_2}$		Z_{01} and Z_{02} depend on transmission-line parameters	

Reflection and Transmission: Norm. Incidence (Continued)

One example is related to standing waves. The superposition of the incident and reflected waves creates a standing-wave pattern in the EM fields. This pattern is described by the **standing-wave ratio**

$$S = \frac{|\tilde{E}_1|_{\text{max}}}{|\tilde{E}_1|_{\text{min}}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{31}$$

Special cases:

- $\eta_1 = \eta_2 \implies \Gamma = 0, \tau = 1, S = 1.$
- If Medium 2 is a perfect conductor, then $\eta_2 = 0 \implies \Gamma = -1, \tau = 0, S = \infty.$

Reflection and Transmission: Norm. Incidence (Continued)

This analogy can also be used to determine the location $\ell_{\max}$ where the electric field intensity in Medium 1 reaches its maximum value. It's the same as finding the locations of the maximum voltages on a transmission line. Here,

$$-z = \ell_{\max} = \frac{\theta_\Gamma + 2n\pi}{2k_1} = \frac{\theta_\Gamma \lambda_1}{4\pi} + \frac{n\lambda_1}{2} \tag{32}$$

for $n = 1, 2, \cdots$ is $\theta_\Gamma < 0$ and $n = 0, 1, 2, \cdots$ if $\theta_\Gamma \geq 0$, where $\lambda_1 = 2\pi/k_1$ and θ_Γ is the phase of Γ.

Power Flow in Lossless Media

The electric and magnetic fields in Medium 1 are

$$\tilde{\mathbf{E}}_1(z) = \tilde{\mathbf{E}}^i(z) + \tilde{\mathbf{E}}^r(z) = \left(E_0^i e^{-jk_1 z} + \Gamma E_0^i e^{jk_1 z}\right)\hat{\mathbf{x}} \tag{33}$$

$$\tilde{\mathbf{H}}_1(z) = \tilde{\mathbf{H}}^i(z) + \tilde{\mathbf{H}}^r(z) = \frac{1}{\eta_1}\left(E_0^i e^{-jk_1 z} - \Gamma E_0^i e^{jk_1 z}\right)\hat{\mathbf{y}}. \tag{34}$$

So, in Medium 1,

$$\begin{aligned}
\tilde{\mathbf{E}}_1(z) \times \tilde{\mathbf{H}}_1^*(z) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ E_0^i e^{-jk_1 z} + \Gamma E_0^i e^{jk_1 z} & 0 & 0 \\ 0 & \frac{1}{\eta_1}\left(E_0^{i*} e^{jk_1 z} - \Gamma^* E_0^{i*} e^{-jk_1 z}\right) & 0 \end{vmatrix} \\
&= \left(E_0^i e^{-jk_1 z} + \Gamma E_0^i e^{jk_1 z}\right)\left(\frac{1}{\eta_1}\left(E_0^{i*} e^{jk_1 z} - \Gamma^* E_0^{i*} e^{-jk_1 z}\right)\right)\hat{\mathbf{z}} \\
&= \frac{1}{\eta_1}\left(\left|E_0^i\right|^2 + \Gamma\left|E_0^i\right|^2 e^{j2k_1 z} - \Gamma^*\left|E_0^i\right|^2 e^{-j2k_1 z} - |\Gamma|^2\left|E_0^i\right|^2\right)\hat{\mathbf{z}} \\
&= \frac{\left|E_0^i\right|^2}{\eta_1}\left(1 - |\Gamma|^2 + \Gamma e^{j2k_1 z} - \Gamma^* e^{-j2k_1 z}\right)\hat{\mathbf{z}}.
\end{aligned} \tag{35}$$

Power Flow in Lossless Media (Continued)

Here,

$$\Gamma^* e^{-j2k_1 z} = \left(\Gamma e^{j2k_1 z}\right)^* . \tag{36}$$

So,

$$\Gamma e^{j2k_1 z} - \Gamma^* e^{-j2k_1 z} = \Gamma e^{j2k_1 z} - \left(\Gamma e^{j2k_1 z}\right)^* = 2j\mathrm{Im}\left\{\Gamma e^{j2k_1 z}\right\} \tag{37}$$

and

$$\tilde{\mathbf{E}}_1(z) \times \tilde{\mathbf{H}}_1^*(z) = \frac{\left|E_0^i\right|^2}{\eta_1}\left(1 - |\Gamma|^2 + 2j\mathrm{Im}\left\{\Gamma e^{j2k_1 z}\right\}\right)\hat{\mathbf{z}}. \tag{38}$$

Therefore, the average power density in Medium 1 is

$$\begin{aligned}
\mathbf{S}_{\mathrm{av1}}(z) &= \frac{1}{2}\mathrm{Re}\left\{\tilde{\mathbf{E}}_1(z) \times \tilde{\mathbf{H}}_1^*(z)\right\} \\
&= \frac{1}{2}\mathrm{Re}\left\{\frac{\left|E_0^i\right|^2}{\eta_1}\left(1 - |\Gamma|^2 + 2j\mathrm{Im}\left\{\Gamma e^{j2k_1 z}\right\}\right)\hat{\mathbf{z}}\right\} \\
&= \frac{\left|E_0^i\right|^2}{2\eta_1}\left(1 - |\Gamma|^2\right)\hat{\mathbf{z}}.
\end{aligned} \tag{39}$$

Power Flow in Lossless Media (Continued)

The term

$$\mathbf{S}_{\text{av}}^{i} = \frac{\left|E_0^i\right|^2}{2\eta_1}\hat{\mathbf{z}} \tag{40}$$

is the average power density of the incident wave and the term

$$\mathbf{S}_{\text{av}}^{r} = -\frac{|\Gamma|^2\left|E_0^i\right|^2}{2\eta_1}\hat{\mathbf{z}} = -|\Gamma|^2\mathbf{S}_{\text{av}}^{i} \tag{41}$$

is the average power density of the reflected wave so that

$$\mathbf{S}_{\text{av1}} = \mathbf{S}_{\text{av}}^{i} + \mathbf{S}_{\text{av}}^{r}. \tag{42}$$

The electric and magnetic fields in Medium 2 are

$$\tilde{\mathbf{E}}_2(z) = \tilde{\mathbf{E}}^t(z) = \tau E_0^i e^{-jk_2 z}\hat{\mathbf{x}} \tag{43}$$

$$\tilde{\mathbf{H}}_2(z) = \tilde{\mathbf{H}}^t(z) = \frac{\tau E_0^i}{\eta_2} e^{-jk_2 z}\hat{\mathbf{y}}. \tag{44}$$

Power Flow in Lossless Media (Continued)

So,

$$\tilde{\mathbf{E}}_2(z) \times \tilde{\mathbf{H}}_2^*(z) = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \tau E_0^i e^{-jk_2 z} & 0 & 0 \\ 0 & \frac{\tau^* E_0^{i*}}{\eta_2} e^{jk_2 z} & 0 \end{vmatrix} = \frac{|\tau|^2 \left|E_0^i\right|^2}{\eta_2} \hat{\mathbf{z}}. \tag{45}$$

Therefore, the average power density in Medium 2 is

$$\mathbf{S}_{\mathrm{av2}}(z) = \frac{1}{2} \mathrm{Re}\left\{ \tilde{\mathbf{E}}_2(z) \times \tilde{\mathbf{H}}_2^*(z) \right\} = \frac{|\tau|^2 \left|E_0^i\right|^2}{2\eta_2} \hat{\mathbf{z}}. \tag{46}$$

For lossless media, η_1, η_2, Γ, and τ are real-valued so that $|\Gamma|^2 = \Gamma^2$, $|\tau|^2 = \tau^2$. Also,

$$\frac{\tau^2}{\eta_2} = \frac{1}{\eta_2} \left(\frac{4\eta_2^2}{(\eta_2 + \eta_1)^2} \right) = \frac{4\eta_2}{(\eta_2 + \eta_1)^2} \tag{47}$$

and

$$\begin{aligned}
\frac{1-\Gamma^2}{\eta_1} &= \frac{1}{\eta_1}\left(1 - \frac{(\eta_2-\eta_1)^2}{(\eta_2+\eta_1)^2}\right) \\
&= \frac{1}{\eta_1}\left(\frac{\eta_2^2 + 2\eta_2\eta_1 + \eta_1^2 - \left(\eta_2^2 - 2\eta_2\eta_1 + \eta_1^2\right)^2}{(\eta_2+\eta_1)^2}\right) \\
&= \frac{1}{\eta_1}\left(\frac{4\eta_2\eta_1}{(\eta_2+\eta_1)^2}\right) = \frac{4\eta_2}{(\eta_2+\eta_1)^2} \\
&= \frac{\tau^2}{\eta_2}.
\end{aligned} \tag{48}$$

This means

$$\mathbf{S}_{\text{av}2} = \frac{\tau^2}{\eta_2}\frac{\left|E_0^i\right|^2}{2}\hat{\mathbf{z}} = \left(\frac{1-\Gamma^2}{\eta_1}\right)\frac{\left|E_0^i\right|^2}{2}\hat{\mathbf{z}} = \mathbf{S}_{\text{av}1}, \tag{49}$$

as expected from power conservation.

Power Flow in Lossless Media (Continued)

Example: Yellow light normally incident on a glass surface

A beam of light with wavelength $\lambda = 0.6$ μm shines through air onto a thick glass surface, traveling in the direction normal to the surface. Suppose that the surface is the x-y plane $z = 0$ and that the glass has $\epsilon_r = 2.25$.

Find

1. the positions of the electric field maxima in the air,
2. the standing-wave ratio,
3. and the fraction of incident power transmitted into the glass.

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1. First, the intrinsic impedance of air is

$$\eta_1 = \sqrt{\mu_1/\epsilon_1} = \sqrt{\mu_0/\epsilon_0} \approx 120\pi(\Omega) \tag{50}$$

Power Flow in Lossless Media (Continued)

and the intrinsic impedance of the glass is

$$\eta_2 = \sqrt{\mu_2/\epsilon_2} = \frac{\sqrt{\mu_0/\epsilon_0}}{\sqrt{\epsilon_r}} \approx \frac{120\pi(\Omega)}{\sqrt{2.25}} = 80\pi(\Omega). \tag{51}$$

Therefore,

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \approx \frac{80\pi - 120\pi}{80\pi + 120\pi} = -\frac{40}{200} = -0.2. \tag{52}$$

So,

$$|\Gamma| = 0.2 \ \text{ and } \ \theta_\Gamma = \pi, \tag{53}$$

which means

$$\ell_{\text{max}} = \frac{\theta_\Gamma \lambda_1}{4\pi} + \frac{n\lambda_1}{2} = \frac{\lambda_1}{4} + \frac{n\lambda_1}{2}, \tag{54}$$

where $\lambda_1 = 0.6\ \mu$m is the wavelength of the light and $n = 0, 1, 2, \cdots$ since $\theta_\Gamma \geq 0$.

Power Flow in Lossless Media (Continued)

2. The standing-wave ratio is

$$S = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{1+0.2}{1-0.2} = \frac{1.2}{0.8} = 1.5. \tag{55}$$

3. The fraction of the incident power transmitted into the glass is

$$\begin{aligned} \frac{S_{\text{av2}}}{S^i_{\text{av}}} &= \frac{|\tau|^2 \left|E^i_0\right|^2 /(2\eta_2)}{\left|E^i_0\right|^2 /(2\eta_1)} = |\tau|^2 \frac{\eta_1}{\eta_2} = \left((1-|\Gamma|^2)\frac{\eta_2}{\eta_1}\right)\frac{\eta_1}{\eta_2} \\ &= 1 - |\Gamma|^2 = 1 - 0.04 \\ &= 0.96. \end{aligned} \tag{56}$$

What are some applications in this case?

- windows
- lenses

In a lossy medium, the intrinsic impedance η_c and propagation constant $\gamma = \alpha + j\beta$ are complex-valued.

Medium 1: ϵ_1, μ_1, σ_1,
$\epsilon_{c1} = \epsilon_1 - j\sigma_1/\omega$, $\gamma_1 = \alpha_1 + j\beta_1$

$$\tilde{\mathbf{E}}_1(z) = E_0^i \left(e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}\right) \hat{\mathbf{x}} \qquad (57)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_{c1}} \left(e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}\right) \hat{\mathbf{y}} \qquad (58)$$

Medium 2: ϵ_2, μ_2, σ_2,
$\epsilon_{c2} = \epsilon_2 - j\sigma_2/\omega$, $\gamma_2 = \alpha_2 + j\beta_2$

$$\tilde{\mathbf{E}}_2(z) = \tau E_0^i e^{-\gamma_2 z} \hat{\mathbf{x}} \qquad (59)$$

$$\tilde{\mathbf{H}}_2(z) = \frac{\tau E_0^i}{\eta_{c2}} e^{-\gamma_2 z} \hat{\mathbf{y}} \qquad (60)$$

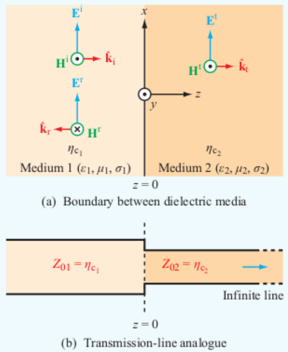


Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Normal Incidence in Lossy Media (Continued)

The reflection and transmission coefficients are

$$\Gamma = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}} \quad \text{and} \quad \tau = 1 + \Gamma = \frac{2\eta_{c2}}{\eta_{c2} + \eta_{c1}}. \tag{61}$$

Example: Normal incidence on a metal surface

Suppose a 1 GHz x-polarized plane wave traveling through the air in the $+z$ direction shines a copper surface located at the x-y plane $z = 0$. The amplitude of the electric field of the incident wave is 12 mV/m and the constitutive parameters for copper are

$$\epsilon_r = \mu_r = 1 \quad \text{and} \quad \sigma = 5.8 \times 10^7 \text{ S/m}. \tag{62}$$

Find the electric and magnetic fields in the air, assuming that the copper is several skin depths thick.

Normal Incidence in Lossy Media (Continued)

Here, Medium 1 is air with

$$\gamma_1 = \alpha_1 + j\beta_1 = jk_1 = j\frac{\omega}{c} = j\frac{2\pi \times 10^9 \text{ rad/s}}{3 \times 10^8 \text{ m/s}}$$

$$= j\frac{20\pi}{3} \text{ rad/m}, \qquad (63)$$

$$\eta_1 = \eta_0 \approx 377\ \Omega, \qquad (64)$$

and

$$\lambda = \frac{2\pi}{k_1} = 0.3 \text{ m}. \qquad (65)$$

For Medium 2, i.e. the copper, $\epsilon_2 = \epsilon_r \epsilon_0 = \epsilon_0$ and

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma/\omega}{\epsilon_2} = \frac{\left(5.8 \times 10^7\right) / \left(2\pi \times 10^9\right)}{1/36\pi \times 10^{-9}}$$

$$\sim 10^9 \gg 1. \qquad (66)$$

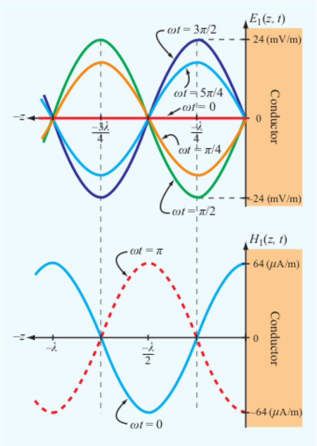


Figure 8-8 Wave patterns for fields $E_1(z, t)$ and $H_1(z, t)$ of Example 8-3.

So,

$$\eta_{c2} \approx (1+j)\sqrt{\frac{\pi f \mu}{\sigma}}$$
$$= (1+j)\sqrt{\frac{(\pi \times 10^9)(4\pi \times 10^{-7})}{5.8 \times 10^7}}$$
$$= 8.25(1+j)\ \text{m}\Omega. \qquad (67)$$

Since $|\eta_{c2}| \ll \eta_1$,

$$\Gamma = \frac{\eta_{c2} - \eta_1}{\eta_{c2} + \eta_1} \approx -1. \qquad (68)$$

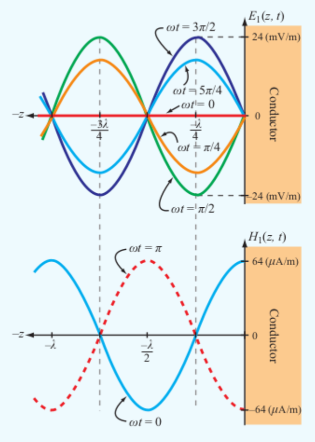


Figure 8-8 Wave patterns for fields $E_1(z,t)$ and $H_1(z,t)$ of Example 8-3.

Normal Incidence in Lossy Media (Continued)

This means

$$
\begin{aligned}
\tilde{\mathbf{E}}_1(z) &= E_0^i \left(e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}\right) \hat{\mathbf{x}} \\
&= E_0^i \left(e^{-jk_1 z} - e^{jk_1 z}\right) \hat{\mathbf{x}} \\
&= -j2E_0^i \sin(k_1 z)\hat{\mathbf{x}} \qquad (69)
\end{aligned}
$$

and

$$
\begin{aligned}
\tilde{\mathbf{H}}_1(z) &= \frac{E_0^i}{\eta_{c1}} \left(e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}\right) \hat{\mathbf{y}} \\
&= \frac{E_0^i}{\eta_0} \left(e^{-jk_1 z} + e^{jk_1 z}\right) \hat{\mathbf{y}} \\
&= \frac{2E_0^i}{\eta_0} \cos(k_1 z)\hat{\mathbf{y}}. \qquad (70)
\end{aligned}
$$

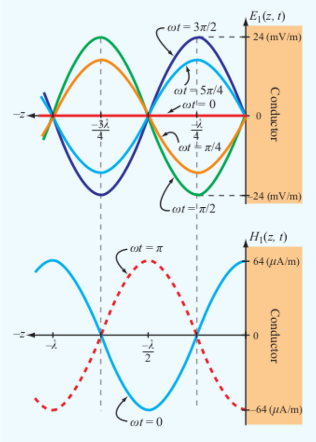


Figure 8-8 Wave patterns for fields $E_1(z,t)$ and $H_1(z,t)$ of Example 8-3.

Therefore,

$$\mathbf{E}_1(z,t) = \text{Re}\left\{\tilde{\mathbf{E}}_1(z)e^{j\omega t}\right\}$$
$$= 2E_0^i \sin(k_1 z)\sin(\omega t)\hat{\mathbf{x}}$$
$$= 24\sin\left(\left(\frac{20\pi}{3}\right)z\right)\sin\left(\left(2\pi\times 10^9\right)t\right)\hat{\mathbf{x}} \quad (\text{mV/m}) \quad (71)$$

and

$$\mathbf{H}_1(z,t) = \text{Re}\left\{\tilde{\mathbf{H}}_1(z)e^{j\omega t}\right\}$$
$$= \frac{2E_0^i}{\eta_0}\cos(k_1 z)\cos(\omega t)\hat{\mathbf{y}}$$
$$= 64\cos\left(\left(\frac{20\pi}{3}\right)z\right)\cos\left(\left(2\pi\times 10^9\right)t\right)\hat{\mathbf{y}} \quad (\mu\text{A/m}). \quad (72)$$

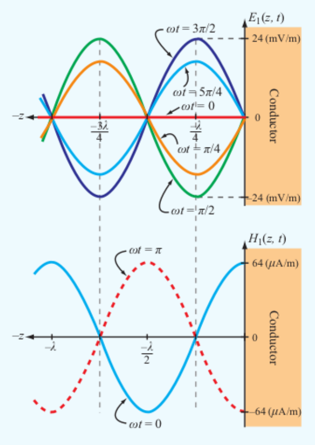


Figure 8-8 Wave patterns for fields $E_1(z,t)$ and $H_1(z,t)$ of Example 8-3.

Normal Incidence in Lossy Media (Continued)

What are some applications?

- mirrors
- shielding
- filters

For **lossless** media, when the incident wave is traveling in a direction that isn't normal to the boundary, it's useful to apply the perspective of **geometric optics**. The rays of the waves **travel in the directions below** and **intersect the boundary at the given angles relative to the normal to the boundary**.

- **incident wave:**
  direction $\hat{\mathbf{k}}_i$, angle $\theta_i \in [0, \pi/2]$,
- **reflected wave:**
  direction $\hat{\mathbf{k}}_r$, angle $\theta_r \in [0, \pi/2]$,
- **transmitted wave:**
  direction $\hat{\mathbf{k}}_t$, angle $\theta_t \in [0, \pi/2]$.

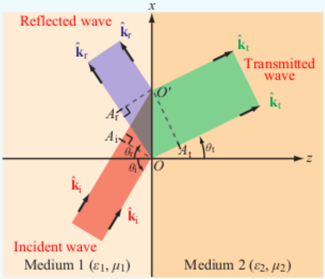


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

Recall that **wavefronts** of a wave are **perpendicular to its rays**.

Snell's Laws (Continued)

Rays of the **incident wave** intersect the boundary at points O and O'.

- The wavefront of the **incident wave** that intersects point O intersects the ray of the incident wave that passes through point O' at point A_i.
- The wavefront of the **reflected wave** that intersects point O' intersects the ray of the reflected wave that passes through point O at point A_r.
- The wavefront of the **transmitted wave** that intersects point O' intersects the ray of the transmitted wave that passes through point O at point A_t.

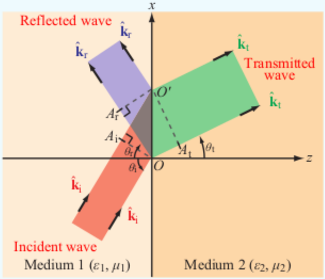


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

Snell's Laws (Continued)

EM waves travel in **Medium 1** at **speed**

$$u_{p1} = 1/\sqrt{\mu_1 \epsilon_1} \qquad (73)$$

and **Medium 2** at **speed**

$$u_{p2} = 1/\sqrt{\mu_2 \epsilon_2}. \qquad (74)$$

The **travel times** for

- the incident wave from A_i to O',
- the reflected wave from O to A_r,
- the transmitted wave from O to A_t

are the same, i.e.

$$\overline{A_i O'}/u_{p1} = \overline{OA_r}/u_{p1} = \overline{OA_t}/u_{p2}. \qquad (75)$$

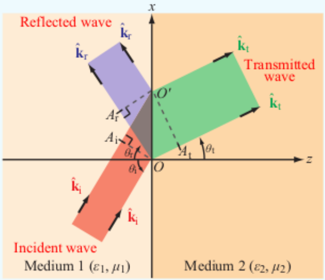
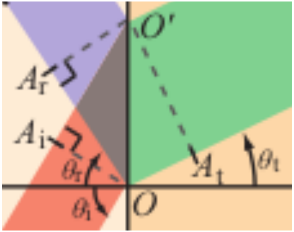


Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

Snell's Laws (Continued)

Notice that this means that $\overline{OA_\text{i}} = \overline{A_\text{i}O'}$.
Also,



$$\overline{A_\text{i}O'} = \overline{OO'}\sin\theta_i \tag{76}$$
$$\overline{OA_\text{r}} = \overline{OO'}\sin\theta_r \tag{77}$$
$$\overline{OA_\text{t}} = \overline{OO'}\sin\theta_t. \tag{78}$$

So,

$$\overline{OO'}\sin\theta_r = \overline{OO'}\sin\theta_i \implies \sin\theta_r = \sin\theta_i \implies \theta_r = \theta_i \tag{79}$$

and

$$\frac{\overline{OA_\text{t}}}{u_\text{p2}} = \frac{\overline{A_\text{i}O'}}{u_\text{p1}} \implies \frac{(\overline{OA_\text{t}})}{(\overline{A_\text{i}O'})} = \frac{u_\text{p2}}{u_\text{p1}} \implies \frac{\sin\theta_t}{\sin\theta_i} = \frac{u_\text{p2}}{u_\text{p1}} = \sqrt{\frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}} = \sqrt{\frac{\mu_{r_1}\epsilon_{r_1}}{\mu_{r_2}\epsilon_{r_2}}}. \tag{80}$$

These are **Snell's laws**.

Homework

Read Chapter 8.

Homework 2 is posted and will be due on Tuesday, May 5, at the beginning of class.

Exam 1 will be Thursday, May 7. Exam 1 review on Thursday.